

ABDUL RAFAY

Fresh Graduate — BS Software Engineering | Full-Stack Web Developer & AI/Agentic AI Enthusiast

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PROFESSIONAL SUMMARY

Motivated Fresh Graduate in BS Software Engineering from Capital University of Science and Technology (CUST) with hands-on experience building full-stack web applications, AI-powered systems, IoT integrations, and blockchain/Web3 platforms. Successfully delivered a Final Year Project (myVISTA — bilingual AI Smart Home Controller) within a 4-person Agile team. Proficient in React, Node.js, Python, Firebase, and REST APIs. Passionate about applying machine learning and agentic AI concepts to real-world problems. Immediately available for full-time roles and internships in Web Development and AI/Agentic AI.

TECHNICAL SKILLS

Languages:	C, C++, Python, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL, Solidity, PHP
Frameworks & Libs:	React.js, React Native, Node.js, Express.js, Vue.js, Expo
AI / ML:	Machine Learning, Sentiment Analysis, Agentic AI, IoT Integration, TensorFlow Lite
Web3 / Blockchain:	Ethereum, Solidity, Hardhat, MetaMask, ERC-721, Web3.js, Sepolia Testnet
Databases:	MongoDB, Firebase Firestore, MySQL, SQL
Cloud & DevOps:	Firebase (Auth, Hosting, Realtime DB), REST APIs, Git, GitHub, CI/CD basics
Tools:	VS Code, XAMPP, Arduino IDE, Postman, Figma, Cloudinary, Hardhat, npm/yarn
Soft Skills:	Agile/Scrum, Problem Solving, Leadership, Communication, Time Management

PROJECT EXPERIENCE

myVISTA – Bilingual AI Smart Home Controller (Final Year Project) ✓ Completed

Sep 2024 – Jun 2025 | React Native · Expo · Firebase · ESP32 · NLP

- Co-developed and successfully delivered a bilingual (English/Urdu) voice-command smart-home controller with gesture controls as FYP within a 4-person Agile engineering team.
- Architected Firebase Realtime Database schema handling 200+ device state events per session with sub-100ms latency.
- Integrated ESP32 microcontrollers via WebSocket for real-time appliance control across 8 device categories.
- Applied Agile sprint planning, weekly standups, and Git branching strategy (feature/dev/main) across a 3-semester timeline.

Smart HomeChef – AI & IoT Smart Cooking System

Jan 2025 – May 2025 | Python · React Native · Firebase · Arduino · TensorFlow Lite

- Engineered an end-to-end AI-powered cooking assistant integrated with IoT sensors (temperature, humidity, gas) via ESP32/Arduino, reducing manual monitoring effort by ~60%.
- Developed a React Native mobile interface with real-time Firebase sync, enabling live sensor dashboards and push notifications for 10+ kitchen events.
- Integrated a TensorFlow Lite ML model achieving 87% accuracy in ingredient recognition from camera input.
- Implemented REST API endpoints in Python/Flask connecting IoT hardware to the mobile frontend.

TB Company – NFT Marketplace (Web3 Full-Stack)

Mar 2025 – Jun 2025 | Solidity · React 18 · Express.js · MongoDB · Cloudinary

- Built a full-stack NFT marketplace on Ethereum Sepolia testnet with MetaMask wallet integration, supporting minting, listing, and trading of ERC-721 tokens.
- Resolved critical MetaMask disconnection bugs and chain-switching errors, reducing transaction failure rate by ~40%.

- Implemented Cloudinary image pipeline for NFT media storage and integrated CoinGecko API for live ETH/USD price charts.
- Designed and deployed 3 Solidity smart contracts (Marketplace, NFTMint, Escrow) verified on Etherscan.

DACVS – Decentralized Academic Credential Verification

Feb 2025 – May 2025 | Solidity · Hardhat · Vue.js · Ethereum Sepolia

- Developed a blockchain-based diploma verification system eliminating certificate forgery with on-chain credential hashing for tamper-proof records.
- Deployed and tested smart contracts using Hardhat on Sepolia testnet; wrote 15+ unit tests achieving 90% contract coverage.
- Built a Vue.js frontend enabling institutions to issue and students to share verifiable credentials via QR code.

DocConnect – Healthcare Appointment Platform (MERN Stack)

Oct 2024 – Jan 2025 | MongoDB · Express.js · React.js · Node.js

- Built a full-stack healthcare platform with appointment booking, role-based auth (JWT), and admin dashboard.
- Designed RESTful API with 20+ endpoints for user management, scheduling, and notification triggers.
- Implemented responsive React.js UI with real-time appointment status updates via polling.

MovieSent – ML Sentiment Analysis System

Aug 2024 – Oct 2024 | Python · Scikit-learn · NLTK · Flask

- Built an ML sentiment classifier trained on 25,000 IMDB reviews achieving 91% accuracy using TF-IDF + Logistic Regression.
- Deployed model as a Flask REST API with a responsive web interface for real-time review analysis.

Personal Portfolio Website

Jan 2025 | HTML5 · CSS3 · JavaScript · SVG Animation

- Designed and developed an 8-page cinematic portfolio site with interactive SVG hero, dual theme system, and smooth page transitions.
- Achieved Lighthouse performance score of 94; integrated downloadable CV and 13 certification assets.

EDUCATION

BS Software Engineering

2021 – 2025

Capital University of Science and Technology (CUST)

CGPA: 2.79 / 4.00 | Graduated 2025 | FYP: myVISTA – Bilingual AI Smart Home Controller (Completed)

ICS (Intermediate in Computer Science)

2020 – 2022

Punjab College Rawalpindi

Result: 60%

Matriculation

2018 – 2020

Honhaar Grammar School

Result: 70%

CERTIFICATIONS & AWARDS

- Participation Certificate — CUST Hackathon 2024 (Capital University of Science and Technology)
- Python for Everybody — University of Michigan / Coursera (Completed 2024)
- Blockchain Fundamentals — Coursera / UC Berkeley (In Progress, 2025)
- React Developer Certificate — Meta Front-End Developer Professional Certificate (In Progress, 2025)

ADDITIONAL INFORMATION

- GitHub: <https://github.com/AbdulRafay> — 7+ public project repositories
- Languages: English (Professional), Urdu (Native)
- Methodologies: Agile / Scrum, Git-flow branching, test-driven development basics

- Availability: Immediately available for full-time roles and internships; open to remote and on-site (Rawalpindi/Islamabad)

References: Available upon request.